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Statement of Purpose

of Kunal Jain (PhD applicant for Fall'25)

As a Research Fellow at Microsoft and previously as an undergraduate researcher at IIIT Hyderabad, I have built a strong determination to pursue a career in research. Over the course of my research journey, I have become fond of the research process irrespective of the outcomes. Particularly, I enjoy diving into uncharted territories and circumventing the unknown with novel ideas. Moving forward, as a PhD student at Georgia Tech, I wish to explore research problems at the intersection of Distributed Systems, Machine Learning and Edge Computing. Machine Learning has made remarkable progress in recent years, primarily driven by increasing sizes of models and datasets. This has led to powerful yet costly and privacy-concerning systems that lack personalization. The increase in mode sizes have led to major concerns: (a) training and inference of models remains limited to the cloud, making them extremely costly for users and (b) training models on cloud raises major privacy and model safety issues as potentially private data needs to be stored globally at the cloud. I believe that both the accessibility and privacy issues can be mitigated by bringing inference and training closer to the edge. My goal is to investigate three critical problems: (i) how can we optimize the edge-cloud division of models during training and inference to ensure both safety and state-of-the-art performance? (ii) Is it possible to personalize current models while maintaining security and privacy? (iii) How can we optimize training and inference at the data center layer in order to make ML solutions available to the next billion users? Overall, I hope to develop systems that balance cutting-edge performance with accessibility, privacy, and personalization.

Background: My research interest was sparked when I took part in the Operational Research Hackathon in December 2020 organized by the state government of Karnataka, India. Our task was to develop an optimal transportation strategy for sending COVID-19 test swabs from collection centers to testing labs. I devised a Mixed Integer Programming solution to minimize costs, along with finding a lower bound on the total allocation expenses. An abridged version of our solution was deployed during the COVID Delta wave in April 2021. Since this formative experience, I have sought out problems which have a broad, positive impact on the world and require deep technical knowledge to solve. This experience exposed me to the process of looking at hard technical problems and learning new concepts to come up with an innovative solution. Furthermore, my TA-ship experience during the period of Nov'21-Dec'22 for three undergraduate and graduate level courses has increased my determination to stay in academia for research work.

Distributed Computing: During my dual degree at IIIT Hyderabad, I tackled the challenge of CV pipeline inference for large-scale camera networks across cities and states. To address this, I proposed a cloud-fog architecture that separated processing and storage layers. The core of my contribution was developing novel edge configurations for CV pipelines, designed to extract maximum information from edge devices while minimizing computational costs and minimizing the network bandwidth requirements for transferring the insights from the edge to the cloud. Our work was presented at the CCGrid 2023 conference, where I gained valuable insights into the importance of scientific communication. The effective presentation and numerous follow-up discussions led to a collaboration with researchers from UNCC. This partnership resulted in ongoing work on algorithmic placement of edge devices on road networks to maximize path coverage, which will be presented at SEC 2024 conference. During the course of this work, I had the freedom to explore and learn how distributed systems, edge devices and systems optimizations have an impact on the real world. It required me to implement different components of the system at different levels (edge, GPUs, networking, distrbuted database, etc) and helped me hone out my technical skills at an early stage of my career. Throughout the process, I had to continuously look for new approaches and learn concepts in order to improve my work, and the explorative nature of research has really motivated me to pursue a PhD.

Efficient Inference for Large Language Models (LLMs): To gain experience on research problems faced by the industry, I joined Microsoft Research as a Research Fellow under the guidance of Dr. Anjaly Parayil. Over the past year, I focused on increasing LLM inference efficiency, a critical issue in today's AI landscape. After thoroughly examining Microsoft's current inference infrastructure and relevant literature, I identified an area of improvement: current approaches focused solely on optimizing scheduling decisions at a single instance and our initial experiments showed that naive load balancing across multiple models can lead to suboptimal end-to-end latency.

This discovery prompted a deep dive into LLM inference internals, exploring how memory and compute interplay affects latency, and analyzing the diverse workloads at Microsoft. Initially, I explored heuristic based approaches to balance workloads but observed marginal gains in LLM inference efficiency. I realized that there was a need for more intelligent methods like RL that provided better solutions to the problem. Upon framing this as a Reinforcement Learning problem, I soon realized two problems: (i) a huge state space and (ii) lengthy experiments requiring faster convergence. By leveraging my systems understanding, I eliminated redundant dimensions and utilized a request mixing study to guide our RL agent, achieving both objectives efficiently and leading to 11% reduction in the average end-to-end latency across different experimental settings.
This project has spawned several follow-up initiatives at Microsoft, which includes a project on optimizing GPU capacity allocation at a data center level while hosting multiple models. We take into consideration low latency and bulk traffic for each model and find an optimal allocation of hardware among them in order to maximize the number of SLOs satisfied. We are focusing on proactive strategies that can help us reduce the cost of running the data center. Other projects include ensuring fairness in LLM inference, and developing novel methods to host Small Language Models (SLMs) on existing infrastructure without compromising LLM services.
This experience has been invaluable, emphasizing the critical importance of maintaining a comprehensive understanding of the field. It has taught me to keep the end-user in focus while identifying problems and developing solutions, and highlighted the crucial role of systems innovation in keeping pace with the latest advances in Machine Learning and Computer Science. These insights have significantly shaped my approach to research and reinforced my passion for tackling complex, real-world challenges in the field of Systems for Machine Learning.

Future Direction: My goal is to bring state-of-the-art ML capabilities closer to the end users, addressing accessibility with systems optimizations. I believe that my experiences in edge device computing, LLM inference, and mathematical modeling and optimization will help me bring the most innovative ML solutions closer to the end-user. Continuous advances in compute reduction, such as quantization, model pruning, and cascading inferences, present an opportunity to bring these advances closer to edge devices. This shift not only aligns with my research goals but also opens up possibilities for model personalization. However, realizing this vision will require significant systems innovation. As we move towards edge-based ML, new challenges will emerge, including the risk of greedy actors seeking disproportionate access to high-quality resources and malicious attempts to extract private data from models. These challenges directly relate to the research questions I proposed earlier: optimizing the edge-cloud division, enabling secure personalization, and optimizing cloud compute utilization. Pursuing a PhD will provide the freedom to explore bold research ideas, an environment of like-minded peers, and opportunities for personal growth.

At Georgia Tech: I want to work with Prof. Alexey Tumanov, Prof. Ada Gavrilovska and Prof. Divya Mahajan at Georgia Tech. I want to focus my research on training and inference in datacenters and on the edge. Dr. Tumanov and Dr. Gavrilovska lead the way on these subjects. Dr. Tumanov work on improving LLM inference systems using creating methods such as chunking and neural architecture search to improve on-device inference highly align with my interests, and the methods in his work often motivate me to think in a different way. Similarly, Dr. Gavrilovska's projects on edge computing and LEO compute clouds also align with my research goals. Her work takes a holistic view of current and emerging hardware stacks and find ways to improve them in interesting manners. Dr. Mahajan's work has spanned from data driven ways of judging the performance of new model-hardware combinations to accelerating training and inference on existing infrastructure, which fits perfectly with my background and will help me achieve my research goals. Overall, their innovative approaches to distributed systems and machine learning present an exceptional opportunity to learn, collaborate, and advance my research in creating accessible, efficient, and privacy-conscious computational systems.

Conclusion: Being the best technical school in the world, I believe Georgia Tech will provide me with the platform and resources I need to excel in my career. A well suited research environment along with the coursework offered during the program will help me become a better independent researcher. I look forward to helping build practical, scalable and safe distributed machine learning platforms in the future. To conclude, I believe my diverse research experience, along with my motivation to learn and grow as a researcher, sets me up well for a PhD at Georgia Tech. I look forward to this next chapter in my life.